



Case Studies

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Objectives

Upon completion of this segment, you will be able to:

- Analyse a case
- Design the strategies for project management
- Prepare a feasibility report

Introduction

In the last segment, we learned about the advantages of using project management software, some characteristics shared by the various types of project management software packages and an illustration of a sample project management package. In this segment, we will examine some case studies related to project management.

After gaining the fundamental knowledge of project management, it is important that we conclude this course in an experiential mode by studying some real-life projects and analysing their project management life cycle, the various opportunities presented in these cases and how some unique threats and challenges were handled by the project management team of these actual projects which were executed in various industries.

This segment provides us with an opportunity to convert the theoretical concepts that we learned into practical project working models. We will also attempt some practical exercises for putting these structured concepts into standardised formats and analyse them. For this purpose, a few practical examples have been provided with generic input data, typical of such projects.



Case Study I: Design and Commissioning of Underground and Elevated Delhi Metro Sections

Scope of the project

The scope of the project included planning, design, tender management, construction supervision, contract management and commissioning of underground and elevated sections of the Delhi Metro.

Strategy for project management

Rail India Technical and Economic Service, also known as RITES, is one of the key constituents of the five-member consortium of general consultants and contributed more than 85% of the total project man-months.

At a wider level, RITES experts, with their skills, supplement the understanding of their counterparts within the consortium on construction, operation and maintenance of railways in India. They also act as a vital communication link in various technical issues between the foreign experts and the client.

RITES deployed about 150 experts (only about 30 expatriates deployed by four foreign partners) in all spheres of activities such as:

- Planning and scheduling
- Design proof checking
- Tender and contract management
- Construction supervision
- Interface management
- System integration
- Trial run, commissioning, and so on.

RITES experts have been able to clearly define the needs and moderate the international requirement in a manner that is best suited to the project and the country as a whole. In short, RITES have competently demonstrated the expertise in delivering a multi-disciplinary project from concept to commissioning.

Man-months input: 9262 (88%) (Including underground and elevated sections).

Initial planning and system design

- Field investigations including topographic survey, geo-technical survey, utility investigation and diversion planning, survey of the existing buildings and structures along the MRTS routes.
- Initial system plans including physical location of stations, detailing of route alignment and passenger data including passenger flow diagram, train operation plan, and so on.
- Initial system designs including alignment design, track layout plan, station layout, rail car depot and inspection facilities, sub-station and power supply systems, current collection, train controls, operation control centre, project estimation, annual budget requirement, cash flow planning and finalisation of the same in consultation with foreign experts.

Tender management (elevated corridors)

- Tender design and tender documentation for various packages of elevated sections (22km of route length) including design approach and criteria; specifications; appointment of



detailed design consultants; coordinating with them for design of viaduct, stations and other service buildings including proof checking of their final design, drawings, bill of quantities and construction details for various packages as proof consultant.

- Preparing tender packages; pre-qualification documents; tender documents; PQ and tender evaluation criteria for the various packages of elevated sections including viaduct, stations and related electrical and mechanical works.
- Pre-qualification of the contractors, invitation of the bids, pre-bid meetings, technical evaluation, financial evaluation, negotiations with the tenderers and recommendation of award of work for rail corridors (elevated sections) contractors.

Tender management (Underground corridor)

- Preparing tender documents (excluding specifications) and prequalification documents for underground section involving about 11 km of twin tunnels.
- Associating with foreign consultants in preparing and finalising the tender design of cut and cover tunnels, underground stations, signalling and telecom system, fare collection system, traction and power distribution system, field instrumentation for monitoring, lifts and escalators and rolling stock as design and construct basis.
- Pre-qualification of tenderers, invitation of bids, pre-bid meetings, receiving offers, technical evaluation and financial evaluation of bids and negotiation and recommendation of award of work for various contract packages involving:
 - underground civil construction,
 - traction and power distribution,
 - signalling and telecom,
 - lifts and escalators,
 - automatic fare collection, and
 - procurement of rolling stock, and so on for the underground section.
- Proof checking of detailed designs for cut and cover, tunnels, signalling and telecom system, power supply and traction system (for underground section), fare collection system and rolling stock as submitted by design and built contractor in association with foreign consultants (partners).
- Coordinating with manufacturers of the rolling stock with respect to proof checking of design giving due consideration for low energy consumption, low operation and maintenance cost, passenger safety, minimum life cycle cost having the facilities of passenger information display system and passenger alarm system and so on.

Construction supervision

- Construction supervision and project management for construction of viaduct, stations, maintenance depots, track work and power supply and distribution system and traction system of various contract packages of rail corridors (elevated section).
- Construction supervision of 11 km of twin tunnels involving:
 - cut and cover section (7 km) using various ground support systems viz. diaphragm wall, sheet piles, ground anchors, soldier piles/mini piles/secant piles, shotcreting and so on; and
 - bored tunnel section (4 km) using laser-guided Tunnel Boring Machines (TBM).
- Supervision of system works comprising of track; traction system; power supply and distribution system; signalling and telecom system using state-of-the-art automatic train protection, automatic train operation, automatic train supervision, and so on; automatic fare collection system using contact-less smart cards/tokens; procurement of rolling stock using regenerative braking, minimum life cycle cost, asynchronous traction motors, high



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- acceleration/deceleration, and so on including their inspection, testing and commissioning for metro corridors (underground section) and rail corridors (elevated section).
- Supervising the work of elevated and underground corridors with respect to high safety standards and construction in an environment-friendly way. This included an initiative in implementing ISO 14001 environment management system in elevated and underground corridors.
- Co-ordination with manufacturer of the rolling stock and other contractors/suppliers for system works for manufacture, installation, testing and commissioning.
- Preparing various manuals including track manual, civil work manual, signalling manual, telecom manual, operating manual, safety manual, rolling stock manual, quality assurance manual and traction manual.

Innovation and complexity of the work

Construction methodology

The Delhi Metro project is the first of its kind in India though the first metro rail system was constructed in Kolkata using conventional methods. It took about 20 years to construct a 16-km corridor and caused severe inconvenience to the public, whereas Delhi Metro has been planned with the latest technology in all fields of construction with meticulous planning and has resulted in on-schedule commissioning of various sections with least inconvenience to the public.

For elevated sections (22km), generally pre-cast, pre-stressed segmental construction has been used to minimise the construction period and inconvenience to public with minimum space requirement on ground. For long spans (60 m), pre-stressed, balanced cantilever construction has been adopted.

The underground corridor traverses under the old walled city of Delhi. A total of 11 km of underground twin metro tunnels have been constructed, out of which about 4 km including Chawri Bazar station have been bored using state-of-the-art laser-guided Tunnel Boring Machines (TBM) and Earth Pressure Balance Machines (EPBM) through rock and soft ground under heavily built-up area. Highly sophisticated tunnel boring machines with record production rate have been used for the first time in India and are considered to be trend setters for future metro rail construction.

Construction of tunnels under thickly populated and commercial hub of Chandni Chowk and Chawri Bazar area of old Delhi encountered a number of unknown wells/ponds which resulted in loss of pressure for TBMs and required plugging/ground stabilisation.

Also, the unexpected type of geological strata varying from sandy silt to quartzite encountered during tunnel boring required major design changes in the cutter head of TBM. The existing buildings within the influence zone of the tunnels were in very poor condition and required condition survey. Based on the same, the buildings were assessed for their expected settlement using a limited tensile strain approach.

Using this data, analysis of the structures was carried out to assess the behaviour of the buildings by the predicted settlement and take necessary protection measures including strengthening of the buildings.

Continuous monitoring with electronic gadgets such as tilt meters, crack gauges and so on was carried out to ascertain the condition of the structures and take additional protection measures if the situation so warranted. Stringent criteria of allowable ground settlement and maximum draw down of 2 m of ground water table, including recharging of the ground water, helped in controlling the



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damage to the existing structures. All this was achieved without losing even a single day of business activities described above.

Limited working space, densely populated commercial area, difficult approach, a lot of existing utilities and running traffic required specialised machinery such as road headers, compact size excavators/loaders, silent generators and so on along with innovative construction methodology involving the use of top-down construction method, secant piles, advance ground probing, soil anchors, self-drilling rock dowels, self-compacting concrete, fibre-reinforced concrete, controlled blasting based on pattern design of explosive and online, real-time monitoring of geo-technical and structural parameters using sophisticated electronic instrumentation.

The balance underground section of about 7 km (including stations, except Chawri Bazar Station) has been constructed by cut and cover method using various ground support methods such as diaphragm wall, sheet piles and soldier piles with lagging and secant piles. Depth of excavation at underground stations varies from 15 to 20 m.

Another innovation of utilising the sunlight in the underground station has been used wherever possible so as to minimise the energy consumption.

Despite the difficulties, site-specific construction methodology and constant monitoring by RITES experts ensured timely completion of the construction. Successful construction of the underground Chawri Bazar Station within a thickly populated and commercial area and Kashmiri Gate underground station across the existing elevated station stands today as a marvel of engineering excellence.

System design

Traction and power supply

Traction and power supply system that has been adopted by the Delhi Metro is energy efficient with complete automation and safety, modern lighting with varying degrees of illumination as per functional requirement and maximum utilisation of daylight and natural ventilation, particularly for underground construction to minimise energy consumption.

Signalling and telecommunication system

Signalling and telecommunication system has been planned using state-of-the-art technologies.

Some of these have been used for the first time in India namely:

- Automatic Train Protection (ATP)
- Automatic Train Operation (ATO) and Automatic Train Signalling (ATS) with combined Operation Control Centre (OCC) along with cab signalling and continuous automatic train control (This enables safe running of trains at 2.5 minutes frequency).
- Trailable point machines
- Coded audio frequency track circuits
- Automatic train supervision and automatic route setting including headway optimisation
- Stainless steel miniature location/junction boxes
- Zero halogen low-smoke cables
- Centralised monitoring and control with data logging
- Non-power SCADA
- Allowing public GSM/CDMA operations in tunnel environment



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The system also provides for terrain mobile radio, closed-circuit TV and state-of-the-art PA/PID system for passenger safety and uses optical-fibre cable transmission.

The Operation Control Centre, a first of its kind, supervises the train operation, monitors passenger information, automatic fare collection system, rail temperature, fire detection and gate control and lifts/escalator functioning.

Automatic fare collection

State-of-the-art automatic fare collection system has been adopted using contact-less smart card and tokens for regular and occasional travellers. Contact-less magnetic tokens have been used for the first time in the world. The system allows 45 passengers/minute per ticket gate and does not permit any ticketless travel.

Rolling stock

Design of rolling stock incorporates highly efficient drive system, high acceleration/deceleration, regenerative braking, stainless steel body, and asynchronous traction motors for minimum life cycle cost. The lightweight coaches are fully vestibuled; air-conditioned; equipped with automatic door closing, Passenger Information Display System (PIDS), and Passenger Alarm (PA) system with automatic audio announcement facility.

Operation and maintenance

With the use of state-of-the-art technology, the system requires reduced Operation and Maintenance (O and M) expenses as compared to Kolkata Metro. Also, only about 45 persons are required per route km for O and M as compared to more than 100 persons per route km of Kolkata Metro.

Passenger safety

The system provides for high standards of passenger safety and includes special walkways throughout the elevated as well as tunnel sections. Evacuation facilities are as per America's safety provisions (NFPA 130) and allows 3 train load of passengers (5000 to 6000) to be evacuated in 3 minutes from underground platform to a safe place (concourse) and in 6 minutes to the ground (outside station). For the elevated section, this time is 4.5 minutes to 7.5 minutes respectively.

Services for handicapped persons

For the first time in India, facilities for handicapped persons have been provided at all stations and in trains in the forms of lifts, ramps, toilets, tactile flooring (guide-way for visually handicapped) and nominated wheelchairs in trains.

High standards of construction

Advance planning of traffic diversion, advance diversion of affected utilities/services, carrying out heavy transportation work during night, barricading the entire construction area from public view to minimise inconvenience to public, close coordination with civic authorities, vigilant construction supervision/monitoring and active participation in implementation of safe construction techniques and ISO 14001 requirements are some of the key factors for establishing high standards of construction.



Achievements

Completion on schedule: Despite the involvement of a number of civic authorities, multi-disciplinary activities and working in congested areas such as Old Delhi and Chawri Bazar, the project has been commissioned ahead of schedule. Line 1 of the project was commissioned 3 months ahead of schedule; the first section of line 2 was commissioned 7 months ahead of schedule whereas the second section of line 2 was also commissioned 3 months ahead of schedule.

Efficient site and traffic management: The revolutionary site management and efficient traffic management at construction sites has been planned in a manner so as to cause minimum inconvenience to the public which has been widely appreciated and is being adopted by other government agencies within the country.

High safety standards: Best safety record during construction. Achieved an accident frequency rate of 0.123 per 1, 00,000 man-hours which is the best in India and is better than most foreign countries.

ISO 14001 certification: RITES experts were involved in getting ISO 14001 certification for construction sites for elevated (22km) and first section of underground corridor (4km) within a record period of 11 months. This has led to the construction in an environment-friendly way.

Timely payment to contractors: 80% in seven days and 100% in 28 days for elevated corridors and 100% in 60 days for international contracts. This has led to effective control at site and timely completion.

Early delivery of coaches: Due to constant proof checking of the design and effective monitoring of production, the delivery schedule of coaches from Korea was advanced by 4 months and was highly appreciated by the client.

Less O and M cost: With the use of state-of-the-art technology, the O and M cost of Delhi Metro is about 60% of that of Kolkata Metro, thus considerable saving to the client.

Value addition

By using modern planning, construction methodologies and management techniques, RITES has set new standards for the construction industry and has shown the way to rapid, safe, environment-friendly and timely construction of such a complex and multi-disciplinary project within the budget and with the least inconvenience to the public. During the execution, the safety standards have been one of the highest in the world with a minimum rate of accidents.

The construction industry in India started following the construction techniques and level of standards introduced by the metro construction, which has since then come a long way.

Benefits to society/economy

- World class metro rail transport with state-of-the-art technology with transfer of technology to India.
- In-house expertise leading to reduced cost of metro construction in other cities of the country.
- Control of air and noise pollution during construction.



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- Water conservation during construction by way of recycling of water and ground water recharging.
- Saving on national resources by segregation and reusing construction waste during construction.
- Increased confidence of funding agencies and government/people due to low accident rate and timely completion.
- Generation of vast employment.
- Opening avenues for the other developing countries for the sale of expertise on metro construction.
- Improvement in international relations.



Case Study II: Design and Manufacturing of Cape Gauge Locomotives and Coaches

Scope of the project

The scope of the project included technical assistance, workshop modernisation, designing, planning, manufacturing, testing, commissioning and field trials of Cape gauge locomotives and coaches in Angola.

Strategy for project management

RITES specialises in providing complete engineering and managerial solutions on a turnkey basis for all transportation problems. It has successfully executed a variety of projects related to design, development, commissioning and field trials of locomotives and coaches in the past in a number of countries. But, RITES had never undertaken the job of designing, developing, manufacturing, testing, commissioning and doing field trials of Cape gauge locomotives and coaches.

For the first time, RITES had made an endeavour in the field of developing Cape gauge rolling stock. The RITES technical team made visits to the premises of the client, understood their requirements, worked out technical details, organised a series of discussions with RDSO, DLW, ICF and BHEL and designed and developed Cape gauge locomotives and coaches. The prototype rolling stock were developed and manufactured as per customer requirements.

Since Cape gauge does not exist over Indian railway, it was, therefore, not possible to conduct field trials of Cape gauge rolling stock in India before their dispatch to Angola. RITES, in association with BHEL, designed and developed the Complete Engine Testing Laboratory (CET Lab) at BHEL for conducting full load tests of the new system. The synchronisation of the power generation and transmission to the traction system along with working of governing system and the control system was tested in the lab.

Working of the complete system was tested in the lab as per actual field conditions. Successful testing of the complete system in the lab gave RITES the confidence about the efficacy of the system developed even in the absence of availability of any facility to conduct actual field trials in India. The Cape gauge locomotives were subsequently supplied to Angola and were commissioned after conducting successful field trials there. The technology has now been successfully proven and established.

After successful commissioning of locomotives and coaches in Angola, RITES had obtained continuous feedback from the customer and on the basis of customer feedback and carried out necessary modifications in the critical sub-systems to ensure their reliability.

RITES offered comprehensive services to CFM, Angola, in the fields of up gradation and management of workshop situated at Lubango. Layout optimisation, material storage system planning, assessment of equipments required for proper maintenance of rolling stock, supply and commissioning of the equipments and other facilities, standardisation of procedures in repairs of rolling stock in the workshop and so on were successfully carried out within the scheduled time frame.

RITES deployed about 20 experts in all spheres of activities such as planning and scheduling, designing, development of prototype, system integration, trial run and commissioning and so on etc.



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RITES experts have been able to clearly define the needs and moderate international requirements in a manner that is best suited to meet the customer requirement.

In short, RITES has competently demonstrated the expertise in delivering a multi-disciplinary project from concept to commissioning.

Initial planning and system design

Preliminary discussions with the customer to access his requirements, visit to site for understanding technical details of the project, accessing the requirement of technical assistance for CFM, assessment of facilities required for modernisation of Lubango workshop, and coordination and discussions with RDSO, DLW, ICF, and BHEL for complete system design and development of prototype of locomotive and coaches including evolution of design of major sub-assemblies like couplers, bogies, traction motors, traction generators, power pack, electrical control system, braking system, governing system, safety protections, etc.

Planning, design, and development of Complete Engine Testing Laboratory (CET Lab) in association with BHEL for testing the system for synchronisation of power generation and transmission as well as functioning of the governing and control system.

Prototype manufacture and project management

1. Finalisation of technical specification of the equipments required for the modernisation of workshop at Lubango, their procurement, co-ordination with vendors for the manufacture of equipments as per customer requirement, inspection and testing of the equipments at vendor's premises, and supply and commissioning of these equipments in Angola.
2. Coordination and project management with DLW for design, development, and prototype manufacture of Cape gauge locomotives.
3. Coordination and project management with ICF for the design, development, and manufacture of prototype Cape gauge coaches.
4. Co-ordination and project management with BHEL for design, development, and prototype manufacture of complete electrical system including traction and control system for rolling stock.
5. Co-ordination with other contractors/suppliers for design and development of major sub-assemblies and other components for their use in the rolling stock system works for manufacture, installation, testing, and commissioning.
6. Co-ordination for design and development of wheel discs to suit track profile and their import for fitment in Cape gauge locomotives and coaches.
7. Preparing various manuals, including maintenance and troubleshooting manuals for Cape gauge locomotives and coaches, driver's instruction manual, safety manual, brake system testing manual, etc.

Innovation and complexity of the work System design

Rolling stock design

The Cape gauge locomotives and coaches have been designed and developed for the first time by RITES to meet the customer requirements. Design of rolling stock incorporates highly efficient power generation system, power transmission system, asynchronous traction motors, suspension and driving system, high acceleration/deceleration, regenerative braking system, and effective governing and control system for minimum life cycle cost. Light weight coaches are fully vestibuled, equipped with Passenger Alarm (PA) system, reliable air brake system for effective braking, and all



passenger amenity fittings including modern lighting with varying degree of illumination as per functional requirement.

Design and development of sub-assemblies

The design and development of sub-assemblies for the rolling stock was a very complex job keeping in view the fact that these sub-assemblies had never been developed in the past for the Cape gauge rolling stock. Their development required complete understanding of the design features of Cape gauge rolling stock. Development of vendors for manufacturing of prototype sub-assemblies, that too for a batch size of very small quantity, and simultaneous confidence building in them was a highly complex job which has been handled successfully on the strength of past record and relationship of this organisation with them.

Design and development of CET Lab

Since Cape gauge does not exist over Indian railway, it was, therefore, not possible to conduct field trials of Cape gauge rolling stock in India before their dispatch to Angola. RITES in association with BHEL designed and developed the Complete Engine Testing Laboratory (CET Lab) at BHEL for conducting full load tests of the complete system. The synchronisation of the power generation and transmission to the traction system along with the working of governing system and control system was tested in the lab.

Working of the complete system was tested in the lab as per actual field conditions. Development of CET Lab and successful testing of the complete system in the lab gave us confidence about the efficacy of the system developed by us even in the absence of availability of any facility to conduct actual field trials in India.

Operation and maintenance

With the use of state-of-the-art technology, the system required reduced O and M expenses as compared to the existing system in Angola. The successful completion of technical assistance programme, including imparting training to CFM personnel, shall accrue to them substantial benefits in the long run by ensuring proper maintenance of rolling stock and thereby increase its availability and productivity.

Passenger safety

Rolling stock supplied by RITES provides for high standards of passenger safety. The audio-visual system provided in the locomotive indicated any abnormality of the system to the driver to enable him or her to take timely preventive action and thereby save any eventuality. The passenger alarm system provided in coaches enabled them to raise an alarm to the driver to stop the train during emergencies. The provision of CBC couplers to prevent parting of trains, anti-telescopic features, air brake system for reliability during brake application, use of fire resistant material and so on are some of the other salient safety features.

Modernisation of the workshop

Comprehensive services in the fields of up gradation and management of workshop situated at Lubango, layout optimisation, material storage system planning, assessment of equipments required for proper maintenance of rolling stock, supply and commissioning of the equipments and other facilities, standardisation of procedures in repairs of rolling stock in the workshop and so on



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were highly complex jobs and required extensive planning and complete reorganisation of the workshop which was successfully carried out within the scheduled timeframe.

Achievements

Timely completion of the programme of technical assistance

The complete programme of technical assistance for imparting training to CFM officials for technical know-how of rolling stock and review and revision of their maintenance systems for safe running of trains was completed well within the scheduled timeframe. This led to the effective management and utilisation of their existing rolling stock assets.

Timely submission of the technical study report for rehabilitation of CFM railway system and maintenance workshops

Timely submission of the technical study report for rehabilitation of CFM railway system and related maintenance workshops enabled them to take proper measures for setting up new standards of maintenance and procurement and commissioning of state-of-the-art machinery and equipment for the proper maintenance of rolling stock.

Timely supply and commissioning of new rolling stock

In spite of the fact that the Cape gauge rolling stock had been designed and developed for the first time in India and no facilities existed for conducting their field tests, RITES not only organised the testing of complete system as per field conditions by developing CET Lab at BHEL, but also ensured timely manufacture of rolling stock as per customer requirement. Complete rolling stock has been delivered and commissioned on time after conducting field trials in Angola.

Timely up gradation of maintenance workshop

Timely provisioning of infrastructure facilities in workshop situated in Lubango, layout optimisation, material storage system planning, assessment of equipments required for proper maintenance of rolling stock, supply and commissioning of the equipments and other facilities, standardisation of procedures in repairs of rolling stock in the workshop and so on benefited the client in revamping its railway system and in introducing the best maintenance practices of the rolling stock. New train services were introduced over CFM Angola on account of better availability of rolling stock.

Less O and M cost

With the use of state-of-the-art technology, the system required reduced O and M expenses as compared to the existing system in Angola. The successful completion of technical assistance programme, including imparting training to CFM personnel resulted in confidence building among CFM, Angola staff in handling their assets efficiently. They had already rehabilitated their assets, which otherwise were lying defunct for long. Substantial benefits in the long run would accrue to them.

Value addition

Imparting technical assistance to CFM, Angola, including training of CFM officials in the field of maintenance and operation of rolling stock, conducting review of their maintenance systems, redefining of existing maintenance practices for safe running of trains and optimisation of the layout of the existing maintenance workshop enabled the client to ensure effective management and



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utilisation of their existing rolling stock assets. Modernisation of maintenance workshop at Lubango resulted in proper upkeep and timely maintenance of rolling stock, thereby regular train services were restored over CFM, which had earlier been discontinued since a long time.

Introduction of new assets in terms of machinery and plant as well as new rolling stock transformed CFM, Angola completely from a non-existent entity to a fully functional railway.

ROI to the client

By engaging RITES for executing the complete project on a turnkey basis, the client accrued by following benefits:

- Easy and quick access to the state-of-the-art technologies in the field of workshop maintenance facilities, resulting in higher efficiency in their working system leading to low operation and maintenance cost and huge savings to the client.
- Access to modern design lightweight train coaches equipped with ultramodern passenger amenities for passenger comfort as well as all safety features related to safe running of the trains.
- Access to the highly efficient Cape gauge locomotives equipped with microprocessor-based state-of-the-art control and diagnostic system.
- Availability of trained manpower for efficient maintenance and management of their costly assets of rolling stock and machinery and plant.
- Road and easy access to the expertise in various fields related to operation and maintenance of rolling stock without incurring a permanent establishment liability, leading to lean client organisation, resulting in better management efficiency.
- Timely identification of the requirements of equipments for modernisation of the maintenance workshop, prompt reorganisation of workshop layout leading to improved productivity and efficiency and optimum utilisation of the resources, early solution to the various difficulties encountered in the re-introduction of new train services, quick and sound decisions and so on.

Benefits to society/economy

The technology for the design and manufacture has now been successfully proven and established. This shall lead to opening up of new avenues for the sale of expertise to the other developing countries and shall enable RITES to establish its credentials in all those countries where the railway system is running on Cape gauge.

Some of the benefits of this project include:

- increased confidence of funding agencies and the government due to timely completion,
- generation of employment,
- improved international relations,
- foreign exchange earnings, and
- improved transport services for Angola.



Case Study III: Strategy for Preparation of Feasibility Report

Introduction

The Golden Quadrilateral of the Indian Railways which connects Delhi, Mumbai, Chennai and Howrah and its two diagonals (Delhi-Chennai and Mumbai-Howrah) constitute about 25% of the broad gauge network of Indian Railways and carry about 55% of the freight carried by the railways. These routes connect centres of production and centres of consumption/trans-shipment.

These routes also carry heavy volumes of passenger traffic which shares the right of way with freight traffic. Due to fixed schedule of priorities, passenger services are accorded priority resulting in heavy detention of freight services. Further, stations are spaced considering the requirements of passenger services whereby freight services are unable to attain an average speed of more than 25 kmph. Line capacity utilisation exceeds 100% in certain sections causing delay to all services. Capacity enhancement works such as improved signalling, addition of lines and electrification are found inadequate to meet the long-term demand.

Freight is being diverted to road, thus causing avoidable incremental loading of roads and consequent direct and indirect economic ill-effects. Against the above backdrop, the government of India decided to implement rail freight corridors operating on a dedicated right of way, connecting Delhi-Mumbai and Delhi-Howrah.

Scope of the project

The assignment proposed for award is the feasibility study for a length of 3,800 route kms which included evaluation of one alternative route which was 1000 kms long. This study was followed by a preliminary engineering survey. This study included assessment of feasibility of DFC, preliminary design of alignment, traffic study, signalling and telecommunication study, study for requirement of rolling stock, study for electrification, assessment of cost and project appraisal.

This feasibility study was intended to serve as the basis for the broad alignment of the DFC and for principles of the operation and design of the DFC.

The study was carried out as per terms of reference communicated by the Nodal Ministry, that is, the Ministry of Railways, through the Railway Board. About 25 experts were handpicked for this project and were drawn from the RITES cadre along with senior officers of the Indian Railways with vast hands-on experience. They were supported by twenty personnel from associated firms.

Terms of reference of this study were as follows:

- To assess the projected growth of freight traffic including containerised traffic on this route considering the existing capacity constraints.
- Transport demand from core sectors including shipping, power, steel and coal during the next 15 years.
- Feasibility of constructing single/double line dedicated freight corridor on one side of the existing double line track between JNPT and Dadri with connections at major identified junctions for transferring of loads and so on.
- DFC along and close to the existing line, where feasible.
- Provision of bye-passes to avoid junction stations.
- Provision of block station at a spacing of 30 to 60 km.
- Optimal length of loop lines appropriate for a 15000 tonne-trailing load.
- Electrification of the lines.



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- Bridges to be fit for 30-tonne axle load and track structure to be fit for 25-tonne axle load.
- Feasibility of induction of special type of wagons for carrying double stack container trains under wire.
- Computerised train control system with GPS-based train-tracking system.
- Construction of new bridges.
- Connectivity and up gradations at Delhi and Mumbai ends.
- Connections avoiding surface crossings by means of rail flyovers and road over/under bridges.
- Assessment of the cost of the project.
- Working out financial and economic rates of return.
- Requirement of rolling stock (wagons and locomotives).

Highlights of the methodology

Based on the base year (2004-05) origin-destination matrix of traffic, a projection of freight traffic for the next 15 years was made, separately identifying the component of container traffic. The number of trains and lines, crossing stations and interchange stations with the existing route and terminals were identified.

Traffic forecast was finalised after extensive consultations with producers, user industries, ports as well as railways. All major developments in infrastructural and industrial growth were taken into account.

The feasibility study has been carried out for construction of the Dedicated Freight Corridor (DFC) on both the routes with connections at major identified junctions for transferring of loads. The lines are designed to carry 25-tonne axle load and bridges will be fit for 30-tonne axle load. Maximum speed is 100 kmph with ruling gradient of 1 in 200. Foot-foot survey was carried out to assess the feasibility of alignment, especially at locations of detour. Field verification was taken up in terms of a number of sections on both corridors.

Existing signalling and telecommunication system and its suitability for DFC, computerised train control system with GPS and requirement of signalling system for level crossing gates and junction stations was studied.

Based on maximum moving dimension, axle load, track loading density, wheel diameter and coupler height, requirement of rolling stock was recommended. Traction system for DFC was based on evaluation of its reliability and economics.

The working expenses and gross earnings have been calculated over the project life for hauling the projected level of traffic. The net earnings, thus arrived at, have been compared with the capital costs comprising construction and rolling stock to arrive at the Financial Internal Rate of Return (FIRR).

Value addition

Tight completion schedule: Despite the involvement of a number of zonal railways, multi-disciplinary activities, and working in the congested areas such as JNPT, Mumbai suburban areas and major towns like Ahmedabad, Vadodara, Bharuch and Palanpur in the Western Corridor and Sonnagar, Mughalsarai, Kanpur and Dadri in the Eastern Corridor and adverse climatic condition during summer and monsoon period, the project was been completed within the scheduled timeframe.



Importance of this study: The total route length for the Golden Quadrilateral and its two diagonals is approximately 10,122 km. These routes annually carry more than 3,000 million passengers and about 55% of the over 600 million tonnes of revenue earning freight traffic lifted by Indian Railways. What is equally important is that these routes serve the core sectors of the Indian economy by carrying raw materials to the plants and finished products to the centres of foreign trade.

Return on investment to client

- Easy and quick access to state-of-the-art technologies in the field of dedicated freight corridor, design of optimised alignment, maintainable track structure and efficient signalling system affording huge potential operational savings to the project.
- Ready and easy access to the expertise in various engineering disciplines/fields and management techniques without incurring a permanent establishment liability leading to lean client organisation, resulting in better management efficiency, early resolution to various construction issues and optimal utilisation of manpower resources.

Benefits to society/economy

- Development of indigenous knowledge of technology leading to steeper learning curve, thus, reducing the cost of planning and implementation of advanced concepts and technologies such as DFC.
- Evolution of experience in project planning and management of mega railway projects which could be utilised in this as well as other major projects.

Summary

In this segment, you examined three case studies, which are examples of successful project management. All three cases are from the Indian project management scenario and illustrate the project management practices of large-sized projects that accrued several benefits to the client as well as the society at large.

The case-based learning is the foundation of the experiential learning methodology where you are encouraged to draw inferences and conclusions from the cases presented to you and make valid managerial judgements by analysing the strengths and weaknesses, if any, of the project discussed in the given case.